A Weekly Demand Briefing for Scotland's Emergency Departments

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AI Mastery Capstone: Industry Integrated AI System Synthesis

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Overview and Industry Context

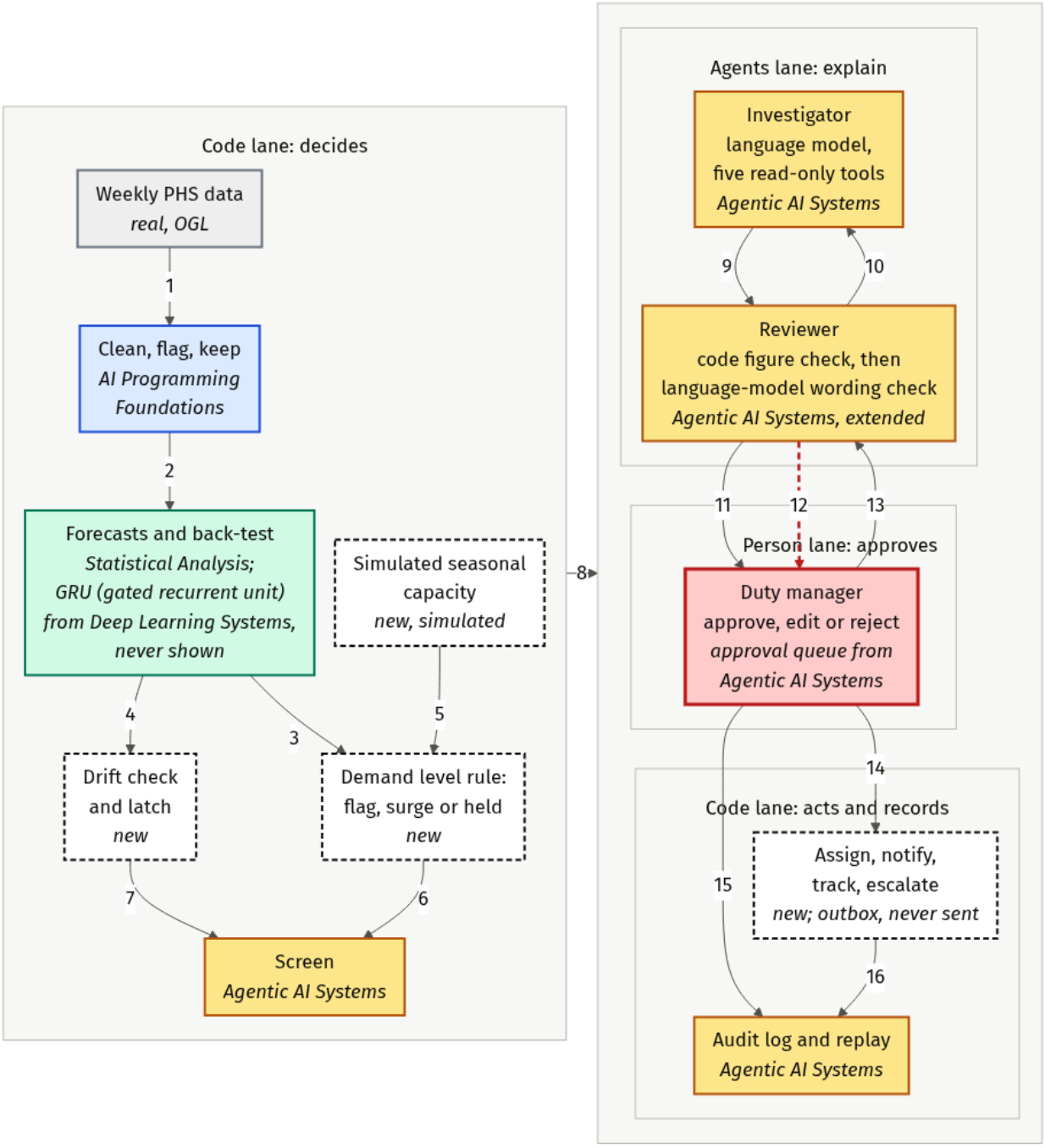
Industry: healthcare operations, unplanned attendances at Scotland's emergency departments (EDs), for the duty managers who staff the front door. Rotas are set weeks ahead: England's National Health Service (NHS) rota code gives resident doctors their rosters at least 6 weeks before a rotation (NHS England, 2025). ED crowding is linked with higher morbidity and mortality (Scottish Government, 2016). A forecast one to four weeks ahead can still inform extra shifts, cover from the staff bank, the board's own pool of flexible staff, or held leave.

At this stage the system gives the duty manager additional information beside their existing process. It replaces no current process and sets no staffing on its own. It needs shadow mode, a run on new weeks with nothing acted on, a pilot with managers who did not build it, and tuning, before it could be a primary operational tool.

Why artificial intelligence (AI)? Simple models forecast demand from open data on accident and emergency (A&E) attendances (Public Health Scotland [PHS], 2026), used under the Open Government Licence (OGL). A gated recurrent unit (GRU), a small neural network, runs beside them only as a comparator (Petravić et al., 2025). A duty manager also needs each forecast explained, with its range and how far to trust it. A language model agent drafts that explanation from read-only tools (Yao et al., 2023), while code sets every number (Figure 1).

Figure 1

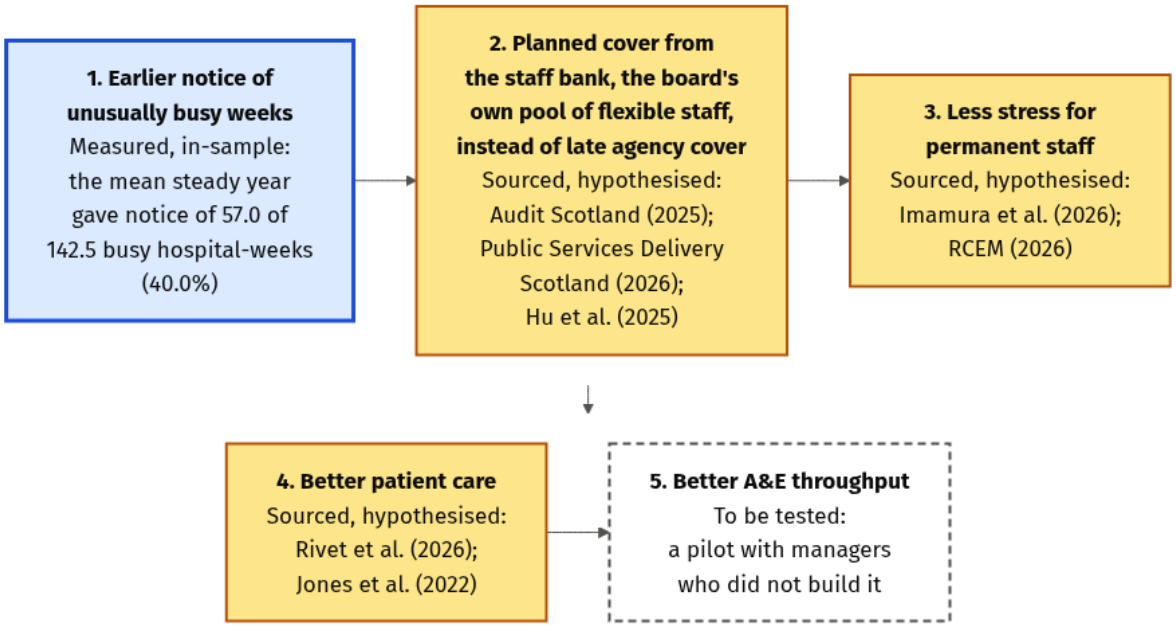
Code Decides, the Agents Explain, a Person Approves



In 2024/25 Scotland spent £256m on agency staff of all kinds and, separately, £353m on the nursing bank (Audit Scotland, 2025, p. 15). Any saving would be mainly the agency premium, with wider value for staff and patients. Figure 2 shows the value chain.

Figure 2

The Value Chain, Each Link Labelled by Its Evidence



Note. Busy means 1.05 times simulated capacity or more; the mean steady year averages 2019 and 2023 to 2025. Sources: Audit Scotland (2025), Public Services Delivery Scotland (2026), Hu et al. (2025), Imamura et al. (2026), the Royal College of Emergency Medicine (2026), shown as RCEM, Rivet et al. (2026) and Jones et al. (2022).

Integration Rationale

I integrated four prior capstone projects (Table 1), spanning data analysis, statistics, deep learning and agentic systems, whose language model brings in generative AI. A fifth, Applied Machine Learning, shaped two decisions: results per site and the pilot.

AI Programming Foundations kept one unexplained merchant's payments, flagged, and the answer changed without them; so the COVID-19 weeks stay in, flagged, as a real shock. In Conduct a Statistical Analysis Using Python ("Statistical Analysis") the changes' standard deviation, 18.63 points, was below either period's spread (22.36 and 21.29), so pairing gave a more precise test. Here a paired interval compares the GRU with the primary on the same weeks. In Deep Learning Systems the best naive forecast placed the day's peak better than either network, so simple models stay the bar to beat. In Agentic AI Systems the Investigator cited figures no tool had returned, and the check caught one in ten; so here code writes every figure and a person approves every brief.

On five replay series at four horizons, outside weeks when a series' forecasts had stopped fitting, the paired interval favoured the primary in 16 of 20 comparisons and the GRU in none. As stated before the run, the GRU stays a comparator. The frontier model, hosted by Anthropic, adds one new technology, its client library, behind Agentic AI Systems' model gateway, chosen by a rule stated before the runs.

Table 1

Prior Capstone Projects and What Each Supplies

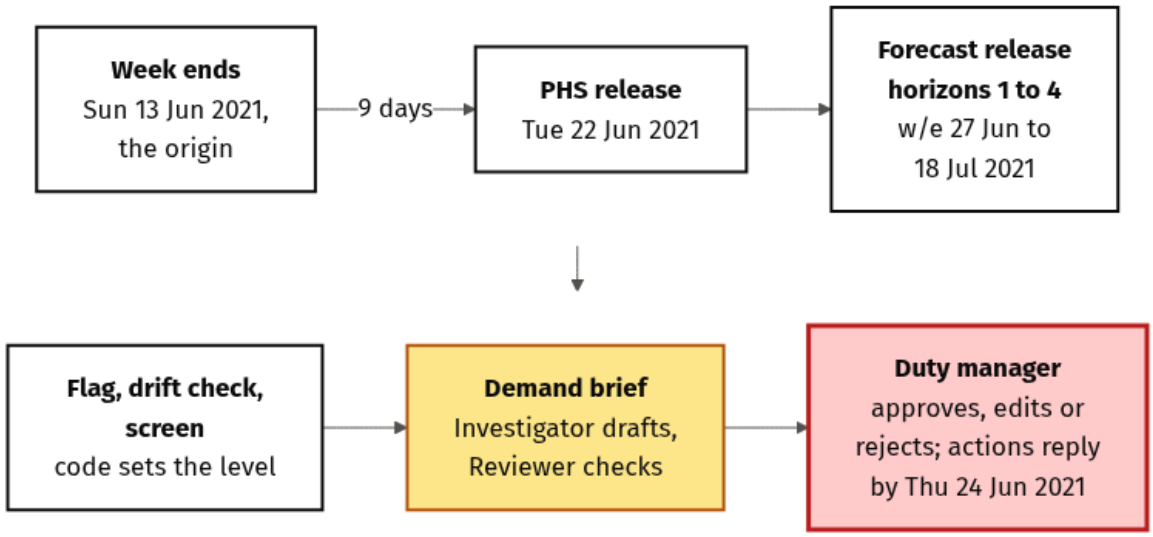
Prior capstone project	What it supplies here
AI Programming Foundations	The weekly load, the COVID-19 weeks kept and flagged
Statistical Analysis	The back-test and bootstrap (Hyndman & Athanasopoulos, 2021, Section 12.5) and the paired test of the GRU
Deep Learning Systems	Seasonal naive, the 4-week moving average and the GRU comparator
Agentic AI Systems	The screen, read-only tools, the figure check, replay, approval and the audit log

System Design and Technical Decisions

Figure 1 shows three lanes: code decides and then acts, the agents explain, and the duty manager approves, edits or rejects each brief. The data are weekly unplanned A&E attendances at 30 hospitals and nationally, 593 weeks to the week ending (w/e) 11 October 2026. Each week is released 9 days after it ends, so the forecasts cover the four weeks after each release, its release horizons (Figure 3).

Figure 3

One Week, From the Tuesday Release to the Duty Manager's Decision



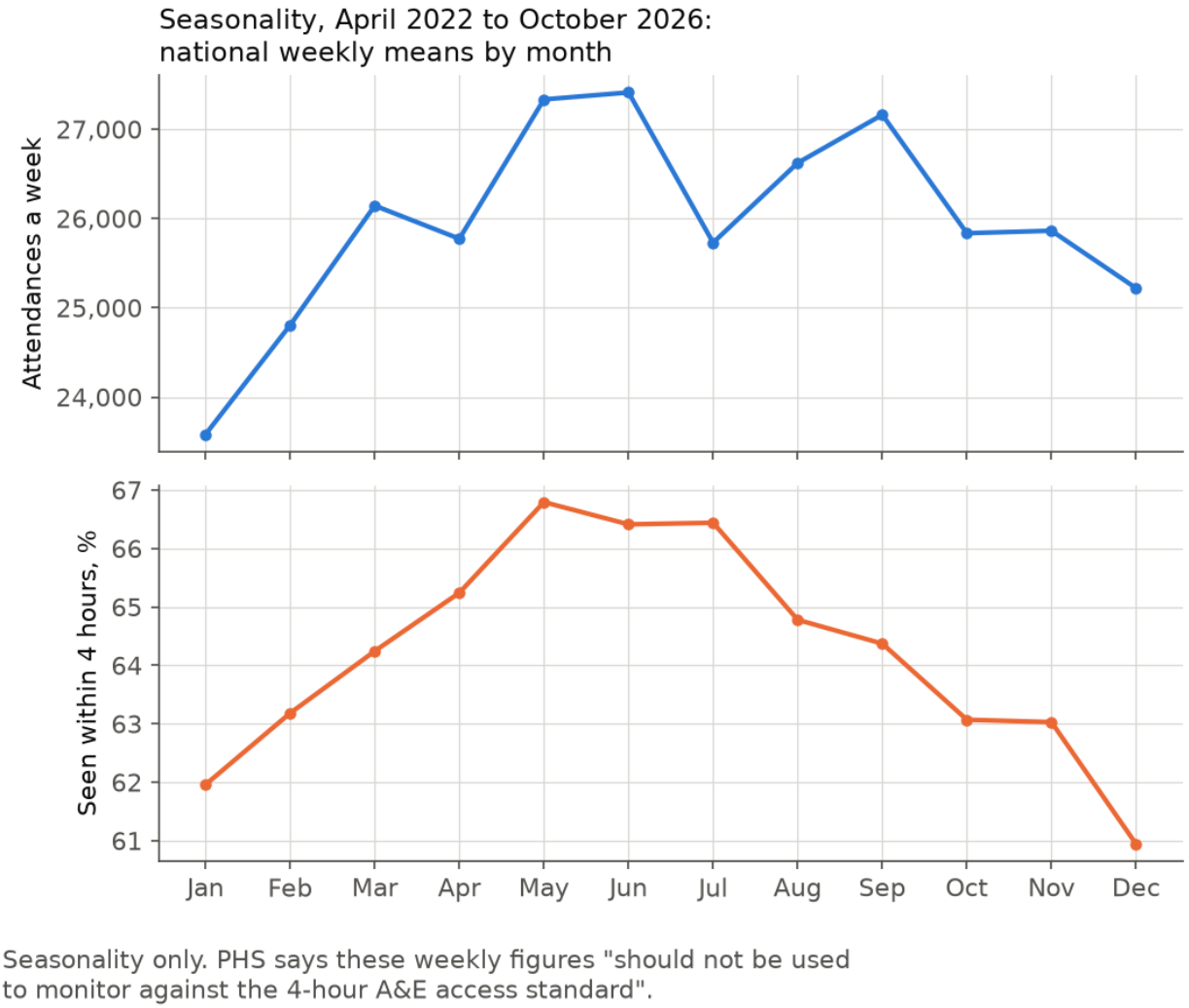
Note. Dates from Ninewells' demand week ending 13 June 2021, whose brief was rejected.

Four simple models forecast each series, among them seasonal naive, the same week last year, a standard benchmark (Hyndman & Athanasopoulos, 2021, Section 5.2). The open data hold no beds or staffing, so I use simulated capacity, a seasonal baseline of past demand: the 90th percentile of the hospital's attendances in the same weeks of the previous three years. Code flags an outlook, one hospital's next four weeks, when the primary model's last 104 errors give

any week at least a 50% chance of reaching 1.05 times capacity. The demand level shows demand unusually high for the time of year; it is not pressure, need or priority. It runs from 1 to 4; a flag raises a week to level 2. Its bands borrow only the structure of England's Operational Pressures Escalation Levels (OPEL) framework (NHS England, 2024). Figure 4 shows the level tracks attendances, not 4-hour performance.

Figure 4

Monthly Attendances Against 4-Hour Performance



Note. Data from PHS (2026), whose data set description holds the quotation in the image.

The drift alarm fires when attendances fall outside the one-week forecast plus or minus 1.96 standard deviations of its recent errors (Hyndman & Athanasopoulos, 2021, Section 5.5) for two released weeks running. It sets the drift latch, a switch that marks the hospital's forecasts untrusted and stops its flag until reset. A surge, an upward breach, takes the latest released week's level. In the design only a person resets the latch; this build simulates each reset. The primary model is then re-selected by its rolling-origin error on the latest 104 released weeks (Hyndman & Athanasopoulos, 2021, Section 5.10). Table 2 gives four tradeoffs.

Table 2

Four Tradeoffs That Shaped the Design

Choice	Gain	Cost
A simple primary model; the GRU only a comparator	A forecast checkable in one sentence	Any pattern the simple models miss
Capacity simulated from past demand	A level tested on real attendances	Illustrative, not about beds
A rare flag	In-sample, steady years outside 2020 to 2022: 53.8% of flags borne out, against 21.2% without one	Only 21.0% of busy outlooks flagged
Hosted Claude Sonnet 5.5, switchable to a local model in Ollama	5 of 7 briefs verified, against none locally	Cost per call, an outside service, no pinned version

The Investigator reasons between tool calls in the ReAct pattern (Yao et al., 2023), writing each figure as a placeholder that code fills. The Reviewer checks every figure against its tool result and screens for the words the Medicines and Healthcare products Regulatory Agency (MHRA) lists as indicative of a medical device (MHRA, 2023, p. 20). A language model then checks the wording. A brief that fails twice is marked "not verified, do not act" and can never be approved. After approval, code sets staffing actions, owners and deadlines, and a language model

words each notification, kept in an outbox, never sent. Figure 5 shows a brief as the duty manager receives it.

Figure 5

A Brief as the Duty Manager Receives It

```
Demand brief: Royal Infirmary of Edinburgh at Little France, week ending 06 Oct
  2019, released Tue 15 Oct 2019
Demand level 2, from the latest released actual (surge).
The demand level shows demand unusually high for the time of year. It is not
  pressure, need or priority.
Level 2 from the surge: the latest released week was 1.05 to under 1.10 times
  simulated capacity, a seasonal baseline of past demand, taken at its lowest over
  the four weeks ahead.
Drift alarm: latched since w/e 29 Sep 2019, after an upward breach; surge: forecasts
  untrusted.
Summary: demand level 2 for Royal Infirmary of Edinburgh at Little France comes from
  the latest released actual, 2,559 attendances (actual, w/e 06 Oct 2019), and not
  from any forecast. The drift latch has been set since w/e 29 Sep 2019 after an
  upward breach, so all forecasts are untrusted until a person approves a latch
  reset. Use the released weeks as the guide for staffing.
Expected range: No forecast range can be trusted while the drift latch is set. The
  released weeks ran from 2,417 attendances (actual, w/e 08 Sep 2019) to 2,688
  attendances (actual, w/e 22 Sep 2019) in the latest weeks.
Model agreement: All forecasts are untrusted while the drift latch is set, so no
  forecast is offered as guidance. The back-test error in this brief is historical
  and does not establish current forecast reliability while the latch is set.
Back-test error: linear's back-test error at release horizon 1 is 3.5%.
Track record: This hospital's past outlook flags were borne out 4 times in 4; its
  unflagged outlooks turned busy 34.5% of the time.
Bank holidays ahead: none in the four weeks ahead.
Watch points:
- Compare the released weeks, 2,633 attendances (actual, w/e 29 Sep 2019) and 2,559
  attendances (actual, w/e 06 Oct 2019), with the rota already set from w/e 20 Oct
  2019 to w/e 10 Nov 2019.
- Check whether extra shifts, bank staff or holding leave are needed from w/e 20 Oct
  2019 to w/e 10 Nov 2019, judged on the released actuals.
- Ask for the latch to be reviewed after the next scheduled data release; a person
  approves a latch reset once the documented reset criteria are met.
```

Note. The Royal Infirmary of Edinburgh's surge brief (w/e 6 October 2019), approved with an edit. The repository's examples/ folder holds a template brief, staff notifications and a "not verified" brief.

Ethical Considerations and Responsible AI

Automation bias and transparency. Agentic AI Systems named automation bias, here a manager approving a brief without weighing it (Goddard et al., 2012), as the next risk at its approval step. Against it, every brief shows what the level means, its range or why none can be trusted, and the hospital's track record; the agent's briefs add the models' agreement. The approver, the duty manager, judges each against six criteria written in advance. A person can override every output, following human oversight as a principle, not a legal duty (European Union, 2024, Article 14).

Misuse and status as additional information. As Statistical Analysis warned its counts could be misused to name code breakers, this tool is never for ranking hospitals, justifying cuts or monitoring the 4-hour standard. It never sees patient data. Whether it is a medical device rests on my reading of the MHRA's guidance, so it stays an open risk (MHRA, 2023).

Bias by hospital size and accountability. Seasonal naive's median error is 11.0% at the smallest sites and 5.4% at the largest. Raised shares by site ran from 35.7% of outlooks at the Royal Hospital for Children to none at Glasgow Royal Infirmary, both in the largest size band, so the burden varies by site more than size. As Applied Machine Learning reported results per public body, each brief shows its own site's error, and the level is never read as need. Owners are by role, on the structure of OPEL's actions by level (NHS England, 2024), and a clinical safety reviewer would sign off before any live use.

Evaluation and Reflection

Every figure is in-sample: the design was tuned on the 2015 to 2026 data that scores it. Error is the mean absolute percentage error (MAPE), as Hyndman and Athanasopoulos (2021, Section 5.8) define it. Figure 6 scores the eight goals, set in advance.

Figure 6

Goals G1 to G8, Each Against Its Threshold

Goal	Met if	Result	Verdict
G1 Forecasts a manager can plan with	within 1 point at every release horizon	-0.18 to -0.74 points against the best simple model	**met**
G2 Honest uncertainty	90% to 98%	95.7% to 96.4% of actuals inside	**met**
G3 Shocks are caught	by Tue 31 Mar 2020; at most 1 a year	known Tue 31 Mar 2020; at most 1 alarm a year	**met**
G4 No unchecked number approved	every approved brief passes; no not-verified brief approved; plant caught	5 of 5 approved pass; 2 not verified, neither approved; plant caught	**met**
G5 The agent never sets a level	always the rule's level	7 of 7 briefs written	**holds**
G6 Briefs are useful to a person	6 of the 7 briefs approved with no factual change, and Reader B makes the approver's decision on at least 3 of the 4 they read	5 of 7 (the two demand-week briefs were refused and rejected); Reader B agreed on 4 of 4	**not met**
G7 Raised levels rare in steady years	under 10% in every steady year	over 10% in 3 years, 10.4% to 11.7%	**narrowly missed**
G8 A flag tells a manager something	at least twice the base rate	2.54 times the base rate, steady years	**met**

G1 to G3, G7 and G8 are in-sample, on 2015 to 2026. G4 to G6 are on the frontier set's seven replay briefs. G5 is an invariant, not a test.

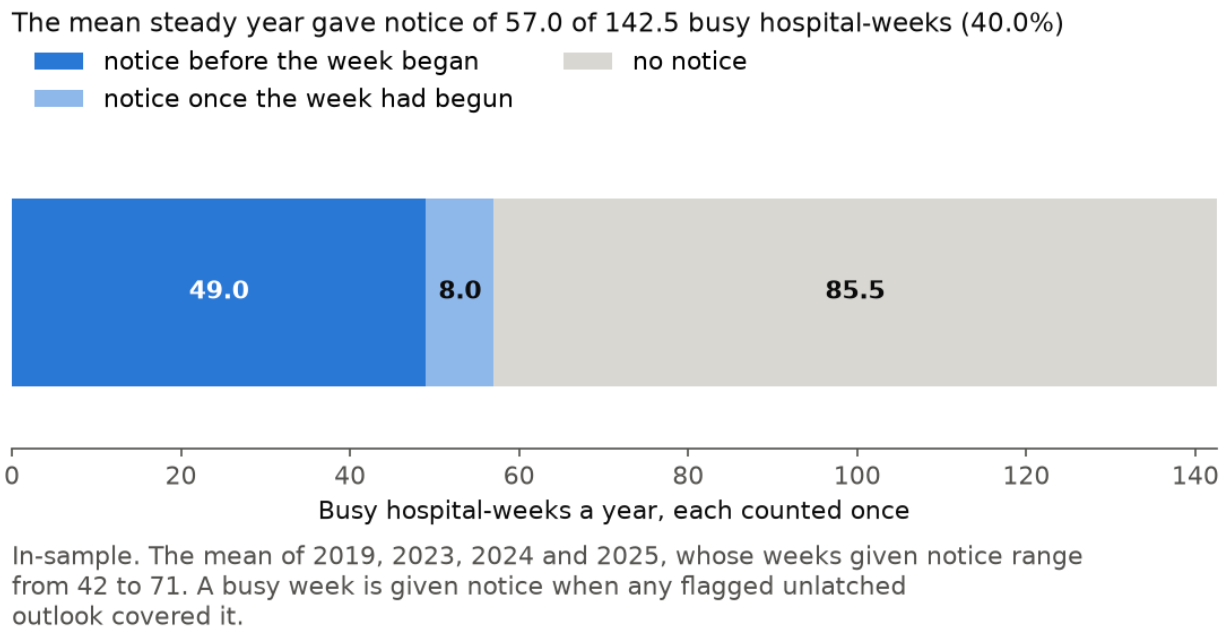
Note. Reader B: a second reader who did not build the system.

What worked. Five goals are met: "forecasts a manager can plan with" (G1), "honest uncertainty" (G2), "shocks are caught" (G3), "a flag tells a manager something" (G8) and "no

unchecked number reaches a manager as approved" (G4). Outside shock weeks, the primary model's national MAPE is within 1 point of the best simple model at every release horizon, and measurably lower only at the fourth. In the mean steady year the flag alone gave notice of 57.0 of 142.5 busy hospital-weeks (Figure 7). "The agent never sets a level" (G5), an invariant, holds.

Figure 7

Busy Hospital-Weeks Given Notice in the Mean Steady Year



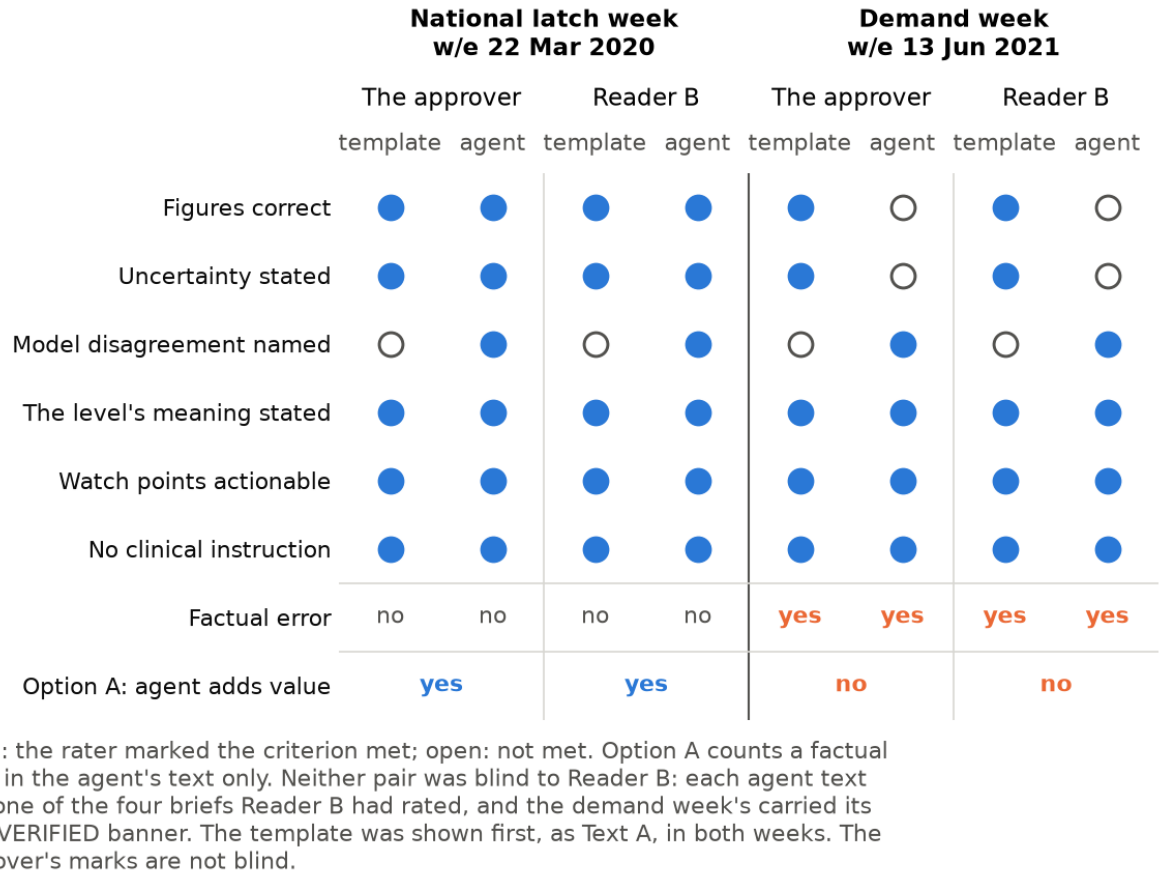
In-sample. The mean of 2019, 2023, 2024 and 2025, whose weeks given notice range from 42 to 71. A busy week is given notice when any flagged unlatched outlook covered it.

What did not work. The adequacy test, a pass mark stated before any brief was written, needed 6 of the 7 replay briefs verified and meeting at least 5 of the 6 criteria. The frontier model came closest, at 5 of 7, and the approver chose its briefs. "Briefs are useful to a person" (G6) had two tests. First, 6 of the 7 briefs had to be approved with no factual change; 5 were, as the two demand-week briefs were refused and rejected. Second, Reader B had to make the approver's decision on at least 3 of 4 briefs, and did on all 4. So G6 is not met; G7 is narrowly missed (Table 3, which lists each failure).

The template comparison, judged by a rule fixed before the ratings (Figure 8), found the agent's explanation added value in the national latch week for both raters: it said every forecast was untrusted while latched, which the template cannot. In the demand week it did not, as that text was the refused brief, whose trend and fit claims the checks had already stopped. So it adds value overall for neither rater.

Figure 8

Template Against Agent: Both Raters, Both Weeks



Filled: the rater marked the criterion met; open: not met. Option A counts a factual error in the agent's text only. Neither pair was blind to Reader B: each agent text was one of the four briefs Reader B had rated, and the demand week's carried its NOT VERIFIED banner. The template was shown first, as Text A, in both weeks. The approver's marks are not blind.

Note. Option A is the rule fixed before the ratings: the agent's text adds value in a week only if it meets every criterion the template's text meets, and at least one more, with no factual error. Raters marked the demand-week agent text's figures wrong because of its NOT VERIFIED banner, though they passed the code check.

Table 3

Failure Cases, With Result and Cause

Run or case	Result	Cause
Local model, round 1	3 of 7 written in time, 0 verified, all refused	Drafts broke the text rules; a fault in my code that calls the model, fixed before round 2, hid each retry's error
Local model, round 2	7 of 7 written, 0 verified	Drafts ran out of output room, thinking included; a retry with thinking off is untried
Frontier model, round 2	5 of 7 verified; both demand-week briefs refused, then rejected	Ninewells' claimed a rising trend from one figure, Lorn & Islands' a flat forecast from one; both echoed code's fit line; the template now writes demand weeks
"Raised levels are rare in steady years" (G7)	10.4% to 11.7% in 2018, 2023 and 2024, against under 10%: narrowly missed; each raise still went to a manager to approve	My untested reading: demand outgrew the years that set capacity; shadow mode tests it
COVID-19, March 2020	Every model missed, one by 152.8%; the latch kept forecasts untrusted	A shock breaks any model trained on past weeks; shadow mode tests the latch
No notice of 85.5 of 142.5 busy hospital-weeks in the mean steady year	The existing process still covers them	A rare flag trades recall for precision; the pilot learns what planners use

Note. The replay briefs are seven real past weeks; the notebook and case study (sections 11 and 13) give each in full.

By that rule, the template now explains demand weeks by default, and the agent latch, surge and model-disagreement weeks. The shipped setting applies this to new live briefs, and template briefs are queued for approval like the agent's. That last ground has no threshold yet, so it keeps no week; setting one is a next step. The template's no-alarm drift line, marked a factual error by both raters, is versioned. The comparison keeps version 1; the shipped template and new

runs say "No drift alarm is recorded; this alone does not establish forecast fit." Recorded runs replay unchanged; shadow mode tests the default.

An AI independent judge read the seven briefs first and drafted the five edits the approver adopted, none of them factual. An AI reviewer, not the system's Reviewer, scored the adequacy test. I built the system and played the approver, which the second reader exists to check.

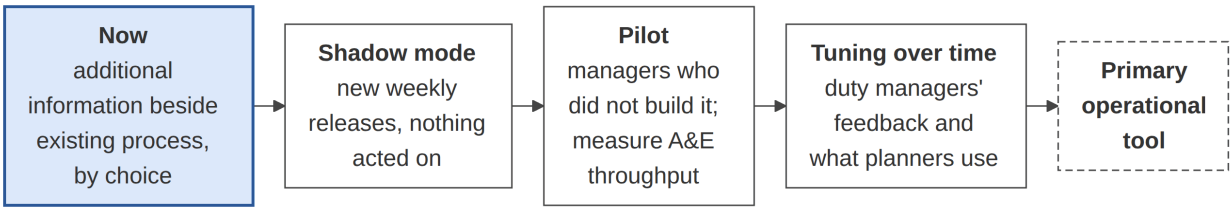
What I learned. Across performance, complexity and responsible use, the GRU's refit and seeds bought no measurable gain, and the frontier model's 5 of 7 verified briefs cost money per call and an outside service. Integration's value lay in the checks between the parts, which stopped both unsupported briefs before approval. The agent's limit showed: it explained latch and surge weeks well enough to approve, but overreached in both demand weeks, where it had to say why demand was high.

Professional Relevance

I kept this a pilot build on purpose, as Applied Machine Learning's first release left its payback for a pilot to show. Figure 9 shows its path to a primary operational tool.

Figure 9

From the Pilot Build to a Primary Operational Tool



This work prepares me for AI solutions architect and machine learning engineer roles, drawing on my enterprise resource planning (ERP) work, where people approve a forecast before anyone acts. Its habits, tests stated in advance, a person at every decision and each failure reported with its cause, are what I would bring to an NHS AI team.

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Glossary: every term used in this paper is defined in glossary.pdf, submitted with it.